

TexCycle AI

AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste

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Research Project

Group ID: R26-IT-081

BSc (Hons) in Information Technology

Specializing in Information Technology

Supervisor: Prof. Dasuni Nawinna

Co-Supervisor: Mr. Nelum Amarasena

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Our Team

Sandeepa S.A.C.

IT22281814

AI Fabric Classification

Siriwardhana N.S.T.N

IT22215338

Environmental Impact Prediction

Madushanka H.M.T.G

IT22221278

Traceability, Recommendations & AI Pricing

Thalagala T.A.D.V

IT22281678

Marketplace Platform & Predictive Analytics

The Problem: Textile Waste in Sri Lanka

Sri Lanka's textile and apparel industry generates considerable waste. Reusable textile material often loses its economic value due to several key inefficiencies:

- Manual sorting and poor waste visibility
- Lack of standardized quality information
- Difficult buyer discovery and unclear pricing
- Limited traceability and sustainability data

TexCycle AI: Our Impact

TexCycle AI aims to transform this challenge into an opportunity by:

- Increasing textile waste reuse
- Supporting circular-economy practices
- Connecting SMEs to affordable materials
- Improving transparency for buyers and sellers
- Enabling data-driven waste management



Shortcomings of Previous Research & Existing Solutions

Existing work has demonstrated individual technologies such as:

- Textile image classification
- Computer vision-based sorting
- Life-cycle assessment & Environmental impact calculation
- Digital traceability
- AI pricing & Recommendation systems
- Textile marketplaces & Predictive analytics

However...

These capabilities are commonly handled as separate solutions, creating significant limitations:

→ **Disconnected Identification & Marketplace**
Fabric analysis does not integrate with sales platforms, hindering efficient transactions.

→ **Sustainability Data Isolation**
Environmental impact metrics are not linked to commercial buying/selling decisions.

→ **Inefficient Buyer Discovery & Pricing**
Finding suitable buyers and establishing fair prices remain manual, negotiation-based processes.

→ **Fragmented Lifecycle Traceability**
Information about a textile's journey is not consistently tied to its transactions.

→ **Unpredictable Market Dynamics**
Demand and sales patterns within the textile waste market are not forecasted.

Our Approach to Address the Gap

TexCycle AI addresses existing inefficiencies by integrating four distinct intelligence layers:

1

Fabric Intelligence
Identifies the textile type and assesses its current condition.

2

Sustainability Intelligence
Evaluates the environmental impact and value of the material.

3

Commercial Intelligence
Determines market worth and suggests potential buyers.

4

Marketplace Intelligence
Predicts listing performance and optimizes for efficient transactions.

Integrated Result: A Circular Journey

These layers combine to create a comprehensive, data-driven workflow for textile waste upcycling:



This integrated four-layer architecture represents the core research contribution of our project.

Constraints & Limitations

- Limited real Sri Lankan textile transaction data available for training.
- Current dataset for predicting environmental impact is synthetic.
- Environmental estimates are highly dependent on accurate source factors.
- AI predictions require rigorous real-world validation to ensure accuracy.
- Market price and demand for upcycled textiles can fluctuate significantly.
- Real-world deployment necessitates continuous model monitoring and adaptation.

Research Question

How can an integrated AI-powered digital marketplace combine textile classification, sustainability assessment, commercial intelligence, and predictive marketplace analytics to support efficient and traceable textile-waste reuse in Sri Lanka?

Sub-questions

Automatic Textile Classification

How can textile waste be automatically classified from images?

Environmental Savings Estimation

How can environmental savings from reuse be estimated?

AI-Powered Pricing & Recommendation

How can AI support textile-waste pricing and buyer recommendation?

Marketplace Behavior Prediction

How can marketplace behavior such as listing success and sales time be predicted?

Integrated Circular Marketplace

How can these capabilities be integrated into one circular marketplace?

Research Gap

Existing Research

Computer Vision

Accurate textile material identification

LCA / Sustainability

Estimation of environmental impacts

Recommendation / Pricing

Support for commercial decisions

Digital Marketplaces

Connection of textile suppliers and buyers

Predictive Analytics

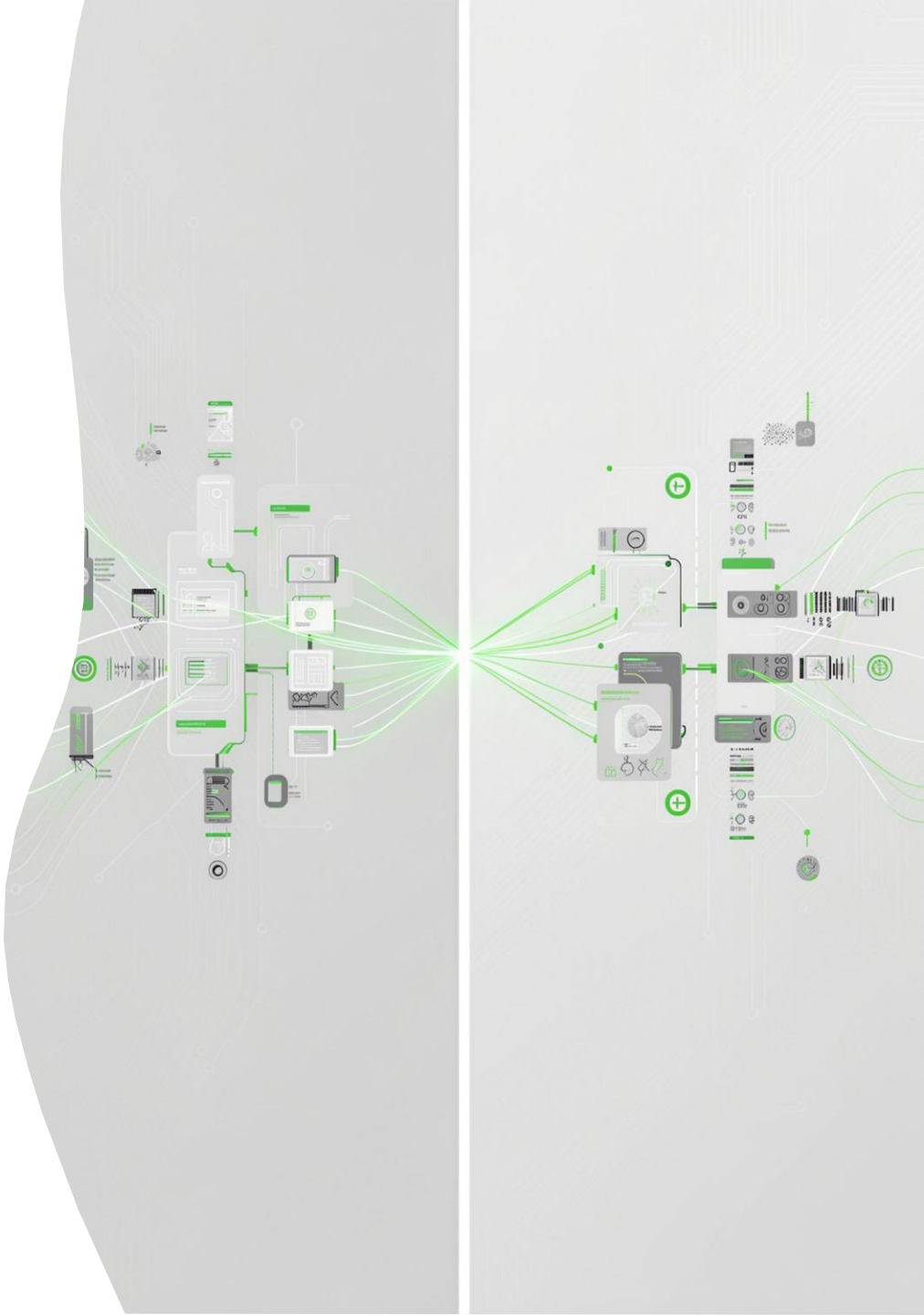
Forecasting of market behavior

Missing Integration

Integrated AI Textile Circularity Platform

Opportunity for a Sri Lanka-focused platform combining:

- Fabric, Sustainability, Commercial, & Marketplace Intelligence
- District-aware information & Local pricing
- Buyer matching & Traceability
- Circular-action recommendations & Marketplace predictions



Four Research Components

Driving one integrated platform



AI Fabric Classification & Condition Intelligence

Sandeepa S.A.C. (IT22281814)

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AI-Based Textile Fabric Classification & Condition Intelligence

Main Objective

This research component leverages advanced AI to meticulously analyze textile waste, providing critical insights into its composition and usability for circular economy applications.

Our solution utilizes an **EfficientNet-B3 16-class classifier** to achieve high-precision textile identification.

Key Outputs

Fabric Type

Precise identification of material composition (e.g., cotton, polyester, blends) using computer vision.

Classification Confidence

A probabilistic score indicating the accuracy of the identified fabric type.

Defect Area

Pinpointing specific areas of damage, stains, or wear on the textile.

Health / Quality Score

An overall assessment of the textile's condition, determining its suitability for various reuse pathways.

Repairability

An evaluation of whether the textile can be repaired and the estimated effort required.

Remaining Lifespan

Prediction of how much usable life is left in the textile, guiding reuse decisions.



The Challenge of Textile Identification & Grading

Current Problem: Uncategorized Waste

- Textile waste arrives as diverse materials, often mixed.
- Variations in fabric types, blends, and colors are common.
- Wide range of quality levels, from pristine to heavily damaged.
- Different degrees of contamination (e.g., oil, dyes, dirt).

Manual Assessment: Inefficient & Inaccurate

- Time-consuming process for large volumes of waste.
- Highly subjective, leading to inconsistent grading.
- Difficult to standardize quality metrics across different assessors.
- Does not scale effectively with increasing waste quantities.

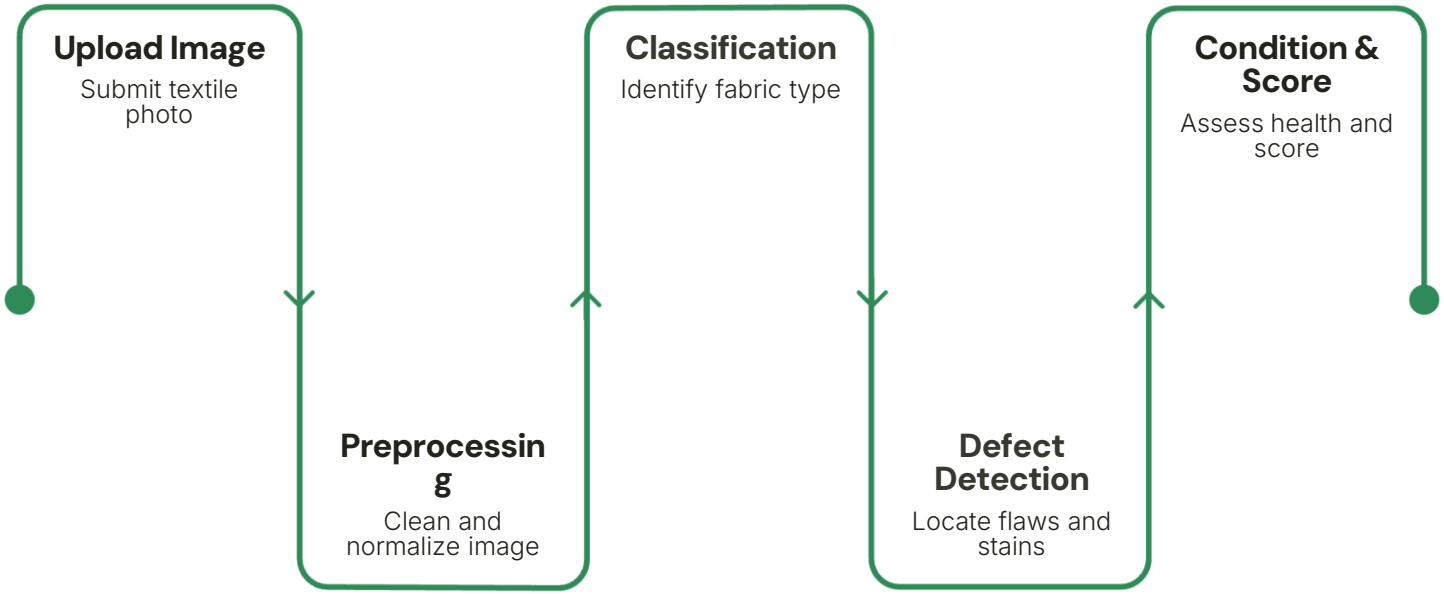
Research Need

There is a critical need for an automated, image-based system that can reliably identify and grade textile waste to facilitate efficient reuse and recycling.

Proposed Solution

AI Fabric Intelligence Pipeline

Our solution employs a sophisticated AI pipeline to comprehensively analyze textile waste, providing detailed insights for effective upcycling.



Underlying Model

At the core of our classification system is an **EfficientNet-B3** model, fine-tuned for textile analysis.

16 Fabric Classes Detected

- Blended
- Chenille
- Corduroy
- Cotton
- Crepe
- Denim
- Fleece
- Leather
- Linen
- Nylon
- Polyester
- Satin
- Silk
- Terrycloth
- Viscose
- Wool

Research Comparison

Existing Approaches & Limitations

Textile Image Classification

Uses computer vision to classify textile/material images.

Limitation: Primarily focuses on material recognition, often lacking condition assessment or marketplace integration.

Textile Defect Detection

Employs computer vision for detecting defects like tears, stains, and wear.

Limitation: Defect analysis is typically siloed and separate from broader marketplace intelligence or sustainability metrics.

Textile Material Datasets

Large datasets support classification and defect detection research, improving model accuracy.

Limitation: Not specifically integrated with Sri Lankan textile-waste marketplace workflows, limiting local applicability and direct impact.

TexCycle AI: Our Integrated System

- Comprehensive Fabric Classification
- Detailed Defect Analysis
- Overall Condition Assessment
- Seamless Marketplace Integration

Latest Technologies Used

Artificial Intelligence & Deep Learning

- CNN
- Transfer Learning
- Computer Vision
- Image preprocessing
- Fabric classification
- Defect detection

Model

EfficientNet-B3

Supporting Technology

- Python
- PyTorch / ML environment
- FastAPI
- REST API
- React frontend

Key Domains

- Computer Vision
- Deep Learning
- Textile Material Science
- Image Processing
- Machine Learning

Evaluation and Validation

Model Evaluation Metrics

Accuracy, Precision, Recall, F1 Score

Standard metrics for comprehensive classification performance assessment.

Confusion Matrix

Detailed breakdown of classification outcomes for each fabric class.

Confidence Analysis

Evaluation of the model's prediction certainty, crucial for real-world application.

Defect Analysis Metrics

- Defect detection quality
- Accuracy of defect-area estimation

Data Utilized

- Proprietary textile image dataset
- Locally collected textile samples from Sri Lanka
- Public textile datasets (where applicable)
- Augmented training images for model robustness

Current Implementation

- Fabric Classifier: `EfficientNet-B3`
- Defect Model: `defect_unet.pth`



Real-World Usage Scenario

Understanding the journey of textile waste through the TexCycle AI platform, from raw input to actionable insights.

Seller Input

- Textile image(s)
- Basic textile information

A seller provides visual data and minimal descriptive details of the textile waste.

AI Processing

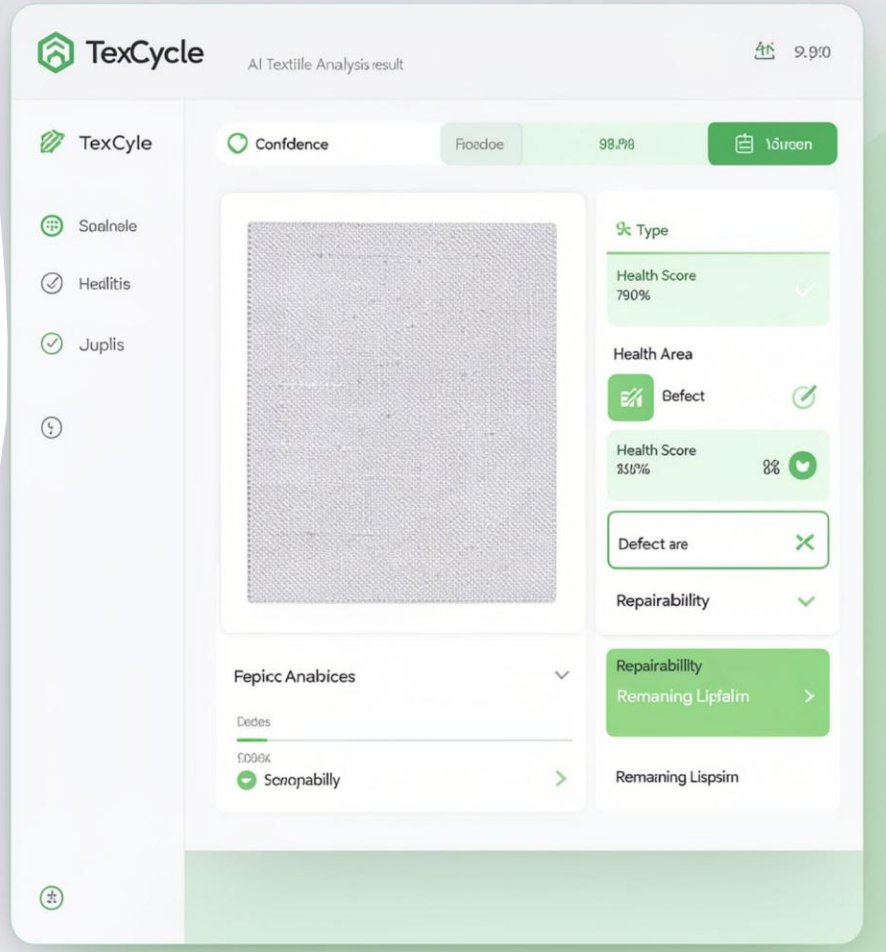
- Identifies fabric type
- Detects visible defects
- Estimates condition & health score
- Estimates repairability & remaining lifespan

Our AI model analyzes the input to generate comprehensive insights.

AI Output

- Fabric: Cotton
- Confidence: 95%
- Health Score: 85
- Defect Area: 5%
- Repairability: High
- Remaining Lifespan: 3 years

Actionable data is provided, enabling informed decisions for reuse and recycling.



Sustainability Intelligence & Environmental Impact Prediction

Siriwardhana N.S.T.N.

IT22215338

Environmental Impact Prediction

This research component focuses on quantifying the positive environmental externalities generated by textile reuse.

Main Question

What measurable environmental value does textile reuse provide within the Sri Lankan context?

Key Outputs

CO_2 Saving
Quantifying the reduction in carbon dioxide emissions achieved through reuse.

Water Saving
Measuring the amount of water conserved by diverting textiles from disposal.

Green Score
A composite index reflecting the overall environmental benefit of a reused textile.

Sustainability Score
An assessment of a textile's contribution to circular economy principles.

Circular-Action Recommendation
Guidance on the most impactful reuse or recycling pathway for a given textile.

Scenario Comparison
Analysis of environmental impacts across different end-of-life scenarios.





The Environmental Cost of Textile Waste

When reusable textile materials are discarded, it leads to a cascade of negative environmental and economic impacts:

Lost Material Value

The inherent value and potential of textile resources are permanently lost, contributing to a linear 'take-make-dispose' economy.

New Raw Material Demand

The continuous production of new textiles requires significant virgin resources, such as water, land, and energy, straining natural ecosystems.

Additional Environmental Impacts

Disposal methods, particularly landfilling, contribute to greenhouse gas emissions and pollution of soil and water bodies.

The Marketplace Gap

Current Problem

Traditional marketplace listings typically only address:

TexCycle AI's Value

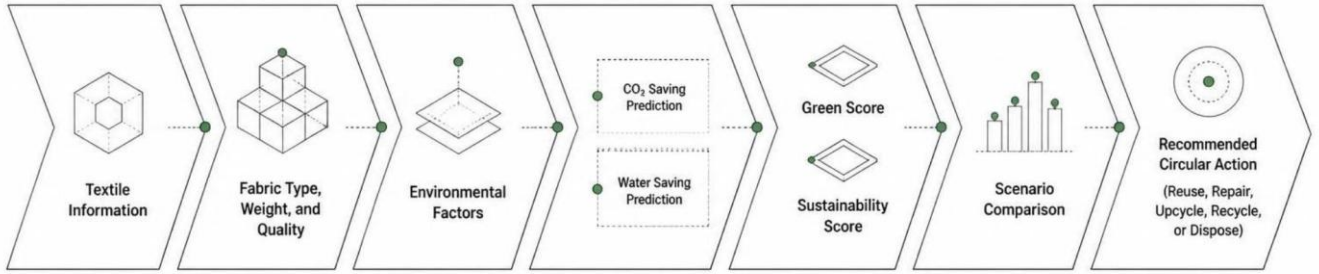
TexCycle AI expands this by additionally asking:

PROPOSED SOLUTION

Sustainability Intelligence Pipeline

Our Sustainability Intelligence Pipeline is designed to transform raw textile information into actionable environmental insights, guiding the most sustainable end-of-life choices for textile waste.

This systematic process ensures that every textile item receives a tailored recommendation, maximizing its environmental value and minimizing waste.



Research Comparison & The Gap

Existing solutions address parts of the textile sustainability challenge, but a critical integration remains unfulfilled.

Research A: Lifecycle Assessment

Provides environmental impact information for textile materials.

Limitation: Usually provides assessment data rather than an interactive marketplace decision layer.

Research B: Textile Sustainability Assessment

Evaluates environmental impacts of textile materials.

Limitation: Not necessarily integrated with individual marketplace listings and buyer/seller decisions.

Research C: Digital Circular Platforms

Provide sustainability/traceability information.

Limitation: Environmental intelligence may not be integrated with all commercial AI components.

Our Proposed System

- Environmental Prediction
- Scenario Comparison
- Explainable Recommendation
- Marketplace Integration

Latest Technologies Used

Our platform leverages cutting-edge technologies and deep domain expertise to deliver robust and insightful textile waste management solutions.

Machine Learning

At the core of our AI capabilities:

- Regression
- Classification
- Random Forest
- Scikit-learn

Sustainability Intelligence

Quantifying environmental impact:

- LCA-based environmental factors
- CO_2 calculations
- Water-saving calculations
- Green Score
- Sustainability Score

Core Technology Stack

The foundation of our platform:

- Python
- FastAPI
- React
- MongoDB
- Machine Learning model pipeline

Key Domains

Interdisciplinary expertise for holistic solutions:

- Life Cycle Assessment
- Environmental Science
- Machine Learning
- Circular Economy
- Sustainability Analytics

Evaluation and Validation

Our rigorous evaluation framework ensures the accuracy and reliability of both our AI models and the environmental impact predictions, providing trustworthy insights for sustainable textile management.

Regression Metrics

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination (R^2)

Classification Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Sustainability Validation

- Environmental-factor verification
- Scenario consistency
- Recommendation validity
- API testing
- Frontend workflow testing

Data Sources

- Life Cycle Assessment (LCA) sources
- Academic literature
- Established sustainability databases
- Textile-specific environmental studies

Current Model Assets

- sustainability_regressor.pkl
- sustainability_grade_model.pkl
- grade_encoder.pkl

Real-World Usage Scenario

To illustrate TexCycle AI's functionality, consider a practical application:



Input

Seller uploads textile details:

- Fabric: Cotton
- Weight: 20 kg
- Condition: Good
- Quality: Medium

Processing

System calculates:

- CO_2 Saving
- Water Saving
- Green Score
- Sustainability Score

Output

Recommended Action:

- Reuse / Upcycle

This process culminates in a comprehensive sustainability analysis, as shown below:

Madushanka H.M.T.G.

Traceability, AI Pricing & Buyer Recommendation

IT22221278

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Research Component: Pricing & Traceability Intelligence

This component focuses on leveraging AI to determine the economic value of textiles and connect them to the right buyers, while also providing a comprehensive digital history.

1

Main Question

What is the textile worth and who should buy it?

2

Key Outputs

- AI price prediction
- Price range
- Buyer recommendation
- Demand-related information
- Listing ranking

3

Enhanced Traceability

- Digital material history

The Problem: Textile Waste Commercialization Challenge

The current textile waste market is fragmented and opaque, creating significant hurdles for both those looking to sell textile waste and those seeking to acquire it for circular economy initiatives.

Sellers Face:

- Unclear resale value
- Manual price negotiation
- Difficulty identifying suitable buyers
- Lack of comprehensive market information
- Limited visibility into material history and impact

Buyers Face:

- Difficult material discovery
- Unclear quality and condition assessment
- Unknown environmental value of materials
- Difficult comparison between diverse textile lots
- Limited seller trust and traceability information

Proposed Solution: AI Commercial Intelligence

Our AI-driven system provides comprehensive commercial intelligence, transforming textile waste into valuable resources by accurately assessing its market worth and connecting it with the most suitable buyers.



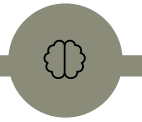
Textile Attributes

Condition + Weight + Fabric Type



Market Information

Real-time supply & demand, historical pricing



AI Price Prediction

- Recommended Price
- Price Range
- Confidence Score



Buyer Recommendation

- Recycler
- Furniture Manufacturer
- Garment Manufacturer
- Fashion Designer
- Insulation Producer
- Cleaning Cloth Producer
- Exporter
- Toy Manufacturer

Research Comparison & The Gap

Our solution synthesizes and expands upon existing research, integrating critical components often found in isolation within the current landscape.

Research A: AI Pricing

Leverages machine-learning models to estimate product or material prices based on various market factors.

- Limitation: Often lacks integration of textile-specific sustainability metrics and detailed buyer segmentation for waste materials.

Research B: Recommendation Systems

Focuses on suggesting products or partners based on attributes, historical behavior, or user preferences.

- Limitation: Traditional systems rarely incorporate nuanced textile condition, environmental impact, or dynamic marketplace demand for waste streams.

Research C: Traceability Platforms

Provides mechanisms to track material movement, origin, and lifecycle stages throughout the supply chain.

- Limitation: Typically operates as a standalone function, disconnected from real-time AI pricing models and intelligent buyer recommendation engines.

Our Proposed System: Bridging the Gap

TexCycle AI integrates these disparate functions into a cohesive platform, offering a holistic solution.

AI Pricing

Accurate, dynamic valuation of textile waste.

Buyer Recommendation

Intelligent matching with suitable circular economy partners.

Traceability

Transparent, digital material history from origin to next use.

Sustainability

Integrated environmental impact assessment and green scores.

Latest Technologies Used

Our platform leverages cutting-edge technologies across multiple domains to deliver intelligent and robust solutions for textile waste management.

1

AI / ML

- Regression
- XGBoost
- Random Forest
- Feature engineering

2

Recommendation

- Attribute matching
- Buyer-category recommendation
- Demand-related ranking

3

Traceability

- Digital material record
- Batch ID
- Transaction history
- Environmental information
- Lifecycle events

4

Technology Stack

- Python
- FastAPI
- Node.js
- React
- MongoDB

5

Key Domains

- Machine Learning
- Recommendation Systems
- E-Commerce
- Supply Chain
- Traceability
- Circular Economy

Evaluation and Validation

Our AI Commercial Intelligence undergoes rigorous evaluation to ensure accuracy, relevance, and data integrity across all its core components, driving trusted insights for the textile waste market.

AI Pricing Metrics

Initial experiment results demonstrate robust performance for our price prediction model:

- **MAE:** 12.24
- **RMSE:** 15.40
- **R^2:** 0.9590

These foundational metrics indicate a strong predictive capability, providing reliable valuation for diverse textile waste categories.

Recommendation Evaluation

We continuously assess the effectiveness of our buyer recommendation engine through key performance indicators to ensure optimal matching:

- Recommendation relevance
- Buyer-match acceptance rate
- Click/selection behavior analysis
- Overall ranking quality and precision

Traceability Evaluation

The integrity and reliability of our traceability system are validated against stringent criteria to ensure a transparent material history:

- Correct batch association across the supply chain
- Comprehensive transaction history records
- Lifecycle-event consistency and accuracy
- Overall data integrity and immutability

Real-World Usage Scenario

Witness TexCycle AI in action: from initial textile upload to intelligent price prediction and optimized buyer matching, creating a transparent circular economy.

Seller Input: Textile Upload

A textile seller uploads details for a batch, specifying criteria like **Fabric: Polyester**, **Quantity: 50 kg**, and **Condition: Good**.

1

AI Processing & Analysis

TexCycle AI instantly processes the data, applying advanced algorithms to evaluate market value, potential applications, and environmental impact.

2

Intelligent Outputs Generated

The system generates a **Predicted Price**, a dynamic **Price Range**, tailored **Buyer Recommendations**, and a comprehensive **Traceability Record**.

3

Optimized Placement Example

For instance, the "Good" condition Polyester might be recommended to a **Recycler** for fiber regeneration or a **Garment Manufacturer** for upcycling.

4

Thalagala T.A.D.V.

Marketplace Platform & Predictive Analytics

IT22281678

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RESEARCH COMPONENT

Research Component: Marketplace Predictive Analytics

Main Question:

How will the textile marketplace perform?

Our research component focuses on developing advanced predictive models to forecast marketplace dynamics and provide actionable insights for all stakeholders.



Listing Success Prediction



Sales-time Prediction



Market-trend Prediction



Demand Analysis



Marketplace Analytics



Predictive Dashboard



Proposed Solution: Marketplace Predictive Analytics

Our advanced AI models analyze historical data and real-time market signals to provide precise predictions, empowering sellers and buyers with crucial market intelligence for the textile reuse and recycling market.

Listing Success Prediction

Forecasts the likelihood of a textile listing being sold, categorizing it as **Sold** or **Unsold**.

Sales Time Prediction

Estimates the **Estimated Days to Sell** for a given textile batch, aiding in inventory and sales strategy.

Market Trend Prediction

Identifies the current market trajectory for specific textile types as **Increasing**, **Stable**, or **Decreasing** in demand or value.

Marketplace Predictive Analytics: Dataset

Our predictive models for marketplace dynamics are developed and demonstrated using a comprehensive, prototype dataset.

Dataset Name: `member4_textile_dataset.csv`

Size: 100,000 records

Features: 36 distinct features capturing various marketplace aspects

Nature: Synthetic/prototype, specifically generated for model development and demonstration.

Commercialization

Driving adoption and value in the TexCycle AI marketplace to foster a thriving circular economy for textiles.

Target Users

Our platform caters to both ends of the textile lifecycle:

Supply Side

- Garment factories
- Textile manufacturers
- Tailors
- Textile waste suppliers
- Apparel SMEs

Demand Side

- Textile recyclers
- Upcycling businesses
- Fashion designers
- Furniture manufacturers
- Garment manufacturers
- Cleaning-cloth producers
- Insulation producers
- Toy manufacturers
- Exporters
- Artisans and SMEs

Platform Value

TexCycle AI delivers distinct benefits to all marketplace participants:

For Sellers

- Better valuation
- Better buyer discovery
- AI-assisted listings
- Market intelligence

For Buyers

- Easier material discovery
- Better comparison
- Flexible purchasing
- Sustainability information
- Seller trust information

Business Model & Commercialization Plan

Our commercial strategy leverages a tiered service model, ensuring accessibility while providing advanced features for specialized needs, complemented by diverse revenue streams to sustain growth and impact.

FREE / BASIC

- Marketplace Access
- Browse textile listings
- Search & filtering
- Basic listing information
- Basic buyer/seller profile

PROFESSIONAL

Includes all Basic features, plus:

- AI recommendations
- Price prediction
- Buyer matching
- Sustainability information
- Marketplace analytics

BUSINESS / ENTERPRISE

Includes all Professional features, plus:

- Bulk procurement
- Advanced analytics
- Demand forecasting
- Digital traceability
- Environmental impact reporting
- Business dashboards
- API / integration services

Potential Revenue Streams

- **Transaction commission:** A small percentage from successful marketplace transactions.
- **Premium subscriptions:** Access to advanced AI analytics for businesses and Professional tier features.
- **Enterprise services:** Tailored solutions for large factories, recyclers, and bulk buyers.
- **API services:** Provision of AI prediction APIs for integration with external organizations' systems.
- **Verified business services:** Enhanced seller/business verification and premium visibility features.

Thank You!

TexCycle AI

Turning Textile Waste into Intelligent Resources

AI + Sustainability + Marketplace + Circular Economy